NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering

Mid-Semester Exam September 2025

Subject: DATABASE MANAGEMENT SYSTEM LAB (~~CS030603~~ CS34/09

Full Marks: 30 **Duration: 2 Hours**

Answer all questions. Make assumption of missing data if any.

Please write precise and to the point answers, irrelevant and lengthy answers may attract penalty.

S. No.		Question	Marks	CO	BL
1.	a.)	Explain three schema architecture.	3	CO1	Remember
	b.)	Differentiate between file management system and database management system.	3	CO1	Understand
	c.)	Explain different constraints in DBMS with example.	4	CO1	Remember
2.		Find the relational algebra, tuple calculus, Domain Calculus and SQL from the following schema → **Emp (eno, fname, lname, dob, supereno, dno, salary,sex, address)** **Dept (dname, dno, mgreno, mgrstartdate)** **Dept_location (dno, dlocation)** **Project (pno, pname, plocation, dno)** **Works_on (eno, pno, hours)**	5*2=10	CO4	Apply
	a.)	Select the employee name, address of employees of "Research" department.			
	b.)	Retrieve details of all employees of department 5 whose salary is between Rs. 30,000 and Rs. 50,000.			
	c.)	Retrieve the project no, project location and employee name who works on "Testing" project.			
	d.)	Retrieve the details of all employees whose address is same as the address of their department.			
	e.)	Select the details of all employees with department name who work on project number 1, 2 or 3.			
3.		Consider the following case study. i.) The university keeps track of each student's name, number, current address and phone, DOB, sex, class. Both personal phone numbers and students' numbers have unique values for each student. ii.) Each department is described by a name, department code, office number and office phone. Department code has unique values for each department. iii.) Each course has a course name, description, course, semester hours, level and offering department code. Course no. is unique in each course. iv.) Each section has an instructor, semester, year, course no. and section no. The section no. distinguishes section of same course that are taught during same semester. v.) A grade report has the student's name, section, and numeric grade. A student is assigned to one department; however, a department can have any number of students. A department can control any number of courses. A course is offered by many number of departments. Draw the ER diagram for this application. Specify the key, relation type and entity type. Note any unspecified requirements and make appropriate assumptions to make the specification complete. Retrieve the relational schemas from the aforesaid application.	7+3	CO2	Apply